

MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Computer Science and Engineering

Faculty of Innovation Engineering

<<Software Project for Course Software Engineering>>

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Amendment History

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1.0	October 2026	D2-Team2	D2-Team2	Initial URS.
2.0	16 October 2026	D2-Team2	D2-Team2	Requirements elicitation, analysis and consolidation.
3.0	18 October 2026	D2-Team2	D2-Team2	Requirements consolidation, source validation and URS revision.
4.0	18 October 2026	(Project Leader)	D2-Team2	Final URS integration and format review; team information and contributions added; captions and references checked.
5.0	18 October 2026	(Project Leader)	( ) Comments received only	Review Comments Received from . Project Leader confirmed three Optional scope decisions and revised requirement clarity and acceptance checks. Final v4 approval remains pending.

Table of Contents

1. Introduction **6**
1.1. Purpose 6
1.2. Notation 6
1.3. Scope6
1.4. Context Diagram 6
1.5. Definitions and Acronyms 7
1.6. References 8
1.7. Overview 8
2. General Description **8**
2.1. System Functions 8
2.2. User Characteristics 8
3. General Constraints **8**
3.1. Software Constraints 8
3.2. Hardware Constraints 9
4. Assumptions and Dependencies **9**
5. Functional Requirements Master List **10**
6. Functional Requirements **11**
6.1. Functional Requirement REQ-1.1 — POI Search and Categories 11
6.2. Functional Requirement REQ-1.2 — POI Details and Opening Information 12
6.3. Functional Requirement REQ-1.3 — Preference Match and Explanation 13
6.4. Functional Requirement REQ-1.4 — Favorites and Add to Trip 14
6.5. Functional Requirement REQ-1.5 — Trip Information Setup 14
6.6. Functional Requirement REQ-1.6 — Automatic Feasible Itinerary 15
6.7. Functional Requirement REQ-1.7 — Unscheduled POIs and Conflict Notices 16
6.8. Functional Requirement REQ-1.8 — Itinerary Editing and Downstream Recalculation 17
6.9. Functional Requirement REQ-1.9 — Deletion and Overwrite Confirmation 18
6.10. Functional Requirement REQ-1.10 — Map and External Navigation 19
6.11. Functional Requirement REQ-1.11 — Macau Start Points and Landmarks 20
6.12. Functional Requirement REQ-1.12 — Budget and Currency Reference 21
6.13. Functional Requirement REQ-1.13 — Travel Bill and Equal Expense Splitting 22
6.14. Functional Requirement REQ-1.14 — Save and Reopen Trips 23
6.15. Functional Requirement REQ-1.15 — Itinerary Plaza and Sharing 23
6.16. Functional Requirement REQ-1.16 — Nearby Food Suggestions 24
6.17. Functional Requirement REQ-1.17 — Relaxed Pace Option 25
7. External Interface Requirement **27**
7.1. Data Interfaces 27
7.2. User Interfaces 27
7.3. Other Interfaces 27
8. Non-Functional Requirements **27**
8.1. System Performance 27
8.2. Information Security 28
8.3. Availability 28
8.4. Capacity 28
9. Domain Requirements **29**
Appendix A Requirements Elicitation and Consolidation **30**
A.1. Source Identification and Collection 30
A.2. Source 1 — Confirmed Customer Needs 30
A.3. Source 2 — Survey Findings 31
A.4. Source 3 — Team-Member Findings 32
A.5. Consolidated Traceability Matrix 32
A.6. Conflicts and Resolution 33
A.7. Evidence Retention and Pending Confirmation 34
Appendix B Scope Decisions **35**
Appendix C — Team Contributions **37**

List of Figures

Figure 1-1 TripMate Macau system context 7

List of Tables

Table 1-1 System context information flows 7
Table 1-2 Definitions and acronyms 7
Table 5-1 Functional requirements master list 10
Table 8-1 System performance requirements 27
Table 8-2 Information security requirements 28
Table 8-3 Availability requirements 28
Table 8-4 Capacity requirements 28
Table 9-1 Macau domain requirements 29
Table A-1 Requirements sources and evidence limits 30
Table A-2 Assigned customer needs and requirement mapping 30
Table A-3 Checked potential user survey findings 31
Table A-4 Reported team member findings 32
Table A-5 Requirements traceability matrix 32
Table A-6 Requirement conflicts and scope resolutions 33
Table B-1 Retained and optional scope 35
Table B-2 Deferred capabilities and MVP alternatives 35

1. Introduction

1.1. Purpose

This User Requirements Specification defines the user-observable behavior and constraints of the TripMate Macau course MVP. It consolidates the existing project baseline with the assigned-customer interview, the potential-user survey and team-member feedback. The requirements guide implementation and acceptance testing.

1.2. Notation

REQ-1.x identifies a functional requirement; PERF, SEC, AVA and CAP identify non-functional requirements; DR identifies a Macau domain rule. H, M and L indicate high, medium and low implementation priority. A shall statement specifies required behavior within its stated scope. An item explicitly marked optional remains optional regardless of its priority. S1, S2 and S3 identify elicitation sources; P identifies the Task 1 project baseline.

1.3. Scope

TripMate Macau is an independent-trip planner for Macau, initially using approximately 30 curated POIs as a project planning assumption. The existing scope covers trip setup, discovery, preference matching, heuristic scheduling, geographic maps, itinerary editing, saved trips and a lightweight Itinerary Plaza. Travel Bill and Equal Expense Splitting, Nearby Food Suggestions and the Relaxed Pace Option are Optional capabilities, not mandatory acceptance functions for the current six-week MVP. The system does not execute bookings or payments.

1.4. Context Diagram

The traveler supplies trip settings, selected POIs, edits and optional expense records, and receives schedules, warnings and maps. Curated POI records support planning. Map tiles and external navigation are provider dependencies; exchange-rate and weather feeds are optional reference inputs. The context diagram shows this boundary and the accompanying table summarizes the information flows.

The system boundary is shown in Figure 1-1; Table 1-1 summarizes the retained data flows.

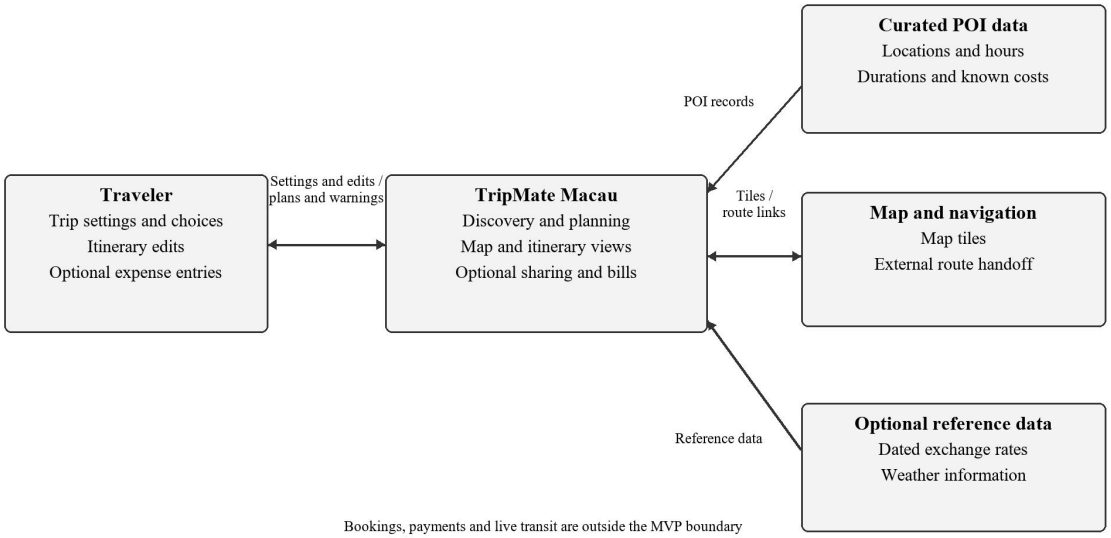


Figure 1-1 TripMate Macau system context

Table 1-1 System context information flows

External entity	Information flow	System
Traveler	Trip settings, selected POIs, edits, expenses →; ← schedules, alerts, maps	TripMate Macau
Curated POI data	Coordinates, hours, categories, durations →	TripMate Macau
Map / navigation provider	Map tiles →; route link ←	TripMate Macau
Optional FX / weather provider	Reference data →	TripMate Macau

1.5. Definitions and Acronyms

Table 1-2 Definitions and acronyms

Term	Definition
URS	User Requirements Specification
POI	Point of Interest, such as an attraction or food stop
MVP	Minimum Viable Product
S1	Assigned-customer interview with ChabataMUST
S2	Potential-user questionnaire and 94 reported responses
S3	Consolidated feedback attributed to four team members as users
P	Final Task 1 Proposal and prior project decisions
MOP / HKD / CNY	Macau pataca / Hong Kong dollar / Chinese yuan

1.6. References

[T2-D] CS250 User Requirements Specifications Description, 29 September 2026. Controlling assignment requirements: three elicitation perspectives and coverage of functional, non-functional and domain requirements.

[T2-T] CS250 User Requirements Specifications Template, 29 September 2026. Official chapter structure and requirement-detail fields.

[S1] TripMate Macau Customer Requirements Interview Original Answers and Expanded Clarifications, four pages. Assigned customer: D2-Team1 (ChabataMUST). Original interview: 10 October 2026. The development team reports confirmation of the expanded wording.

[S2-R] TripMateMacau travel-planning questionnaire platform summary, six pages, 94 reported responses. [S2-Q] Potential-user questionnaire design, four pages. Collection: 13 to 15 October 2026.

[S2-X14], [S2-X15] and [S2-X16] Open-answer exports, 94 records each. Platform Q15 and Q16 correspond to the questionnaire parts Q15.1 on currencies and Q15.2 on splitting expenses.

[S2-S] Potential-user requirements synthesis, eight pages. This is a secondary interpretation; its numerical engineering proposals require separate agreement.

[S3] Team Members as Users consolidated summary version 2, 13 October 2026, reporting feedback attributed to Members 1 to 4.

[P] D2-Team2 TripMate Macau Project Proposal Final, 8 October 2026. Project scope and prior design decisions; elicited Task 2 needs take precedence where they differ.

1.7. Overview

Sections 2–4 provide context and constraints; Sections 5–6 define 17 functions; Sections 7–9 describe interfaces, quality and domain rules; appendices provide traceability and exclusions.

2. General Description

2.1. System Functions

Four home shortcuts provide Trip Planning, Favorites, My Macau Trips and Itinerary Plaza. Four persistent bottom tabs provide Home, Discover, Map and Mine. The core flow is to enter trip details, select POIs, generate a candidate itinerary, inspect or edit it, and save or share it. Optional bills are accessed from an itinerary. The planner, map and saved itinerary shall use the same stop order after each accepted change.

2.2. User Characteristics

Independent visitors, first-time Macau travelers, short-stay tourists and small friend groups. Self-service mobile and desktop browser users; no specialist training required.

3. General Constraints

3.1. Software Constraints

The working technical plan uses Vue 3 and Tailwind CSS for the front end, a JavaScript or TypeScript heuristic planner, Leaflet with OpenStreetMap tiles, and lightweight JSON or SQLite persistence or an equivalent implementation. These are implementation assumptions from the existing draft, rather than

customer-mandated products. External services depend on provider availability and usage terms. The delivered behavior shall retain a usable fallback when an optional provider fails.

3.2. Hardware Constraints

Responsive browser on standard phones and computers; no dedicated hardware. GPS/location permissions are optional; manual Macau origin is available.

4. Assumptions and Dependencies

The draft assumes a six-week course MVP and an initial curated dataset of approximately 30 POIs. These are project planning assumptions, not verified customer capacity thresholds. Opening times, transport durations and costs may be incomplete or estimated and shall be identified accordingly. Navigation and optional exchange-rate or weather services may be unavailable. No numerical response-time, accuracy, concurrency or uptime guarantee has been agreed in the supplied customer evidence. Weather information is for reference; automatic weather-aware replanning remains deferred.

5. Functional Requirements Master List

MVP Status is independent of Priority (H/M/L). Core is required for this six-week MVP; Optional retains its definition and design and is implemented only if time permits; Deferred is outside this MVP. Table B-1 records the scope basis.

Table 5-1 Functional requirements master list

Req. ID	Requirement name	Requirement description	Priority	MVP Status	Sources
REQ-1.1	POI Search and Categories	Find Macau POIs by keyword and category	H	Core	S1,S2,S3,P
REQ-1.2	POI Details and Opening Information	Show practical visit information	H	Core	S1,S2,S3
REQ-1.3	Preference Match and Explanation	Show preference-based match percentage and short reason	M	Core	S3,P
REQ-1.4	Favorites and Add to Trip	Separate saving a POI from selecting it for a trip	H	Core	S1,S2,S3,P
REQ-1.5	Trip Information Setup	Capture trip constraints before planning	H	Core	S1,S3,P
REQ-1.6	Automatic Feasible Itinerary	Generate ordered daily POI schedule minimizing unnecessary backtracking	H	Core	S1,S2,S3,P
REQ-1.7	Unscheduled POIs and Conflict Notices	Never silently discard selected POIs	H	Core	S1,S3
REQ-1.8	Itinerary Editing and Downstream Recalculation	Edit visit times and transport while rechecking later stops	H	Core	S1,S2,S3,P
REQ-1.9	Deletion and Overwrite Confirmation	Confirm destructive itinerary actions	H	Core	S3,P
REQ-1.10	Map and External Navigation	Visualize itinerary route and allow leg-level navigation	H	Core	S1,S2,S3,P
REQ-1.11	Macau Start Points and Landmarks	Offer common Macau entry points and manual start	H	Core	S1,S3,P
REQ-1.12	Budget and Currency Reference	Show budget and MOP/HKD/CNY reference conversions	M	Core	S1,S2,S3,P
REQ-1.13	Travel Bill and Equal Expense Splitting	Record optional trip expenses and compute equal shares.	M	Optional	S3, P
REQ-1.14	Save and Reopen Trips	Allow retrieval of previously generated and edited trips	H	Core	S3,P
REQ-1.15	Itinerary Plaza and Sharing	Browse and voluntarily publish Macau itineraries	M	Core	S1,S3,P
REQ-1.16	Nearby Food Suggestions	Suggest curated food POIs near selected stops.	M	Optional	S2, S3
REQ-1.17	Relaxed Pace Option	Offer simpler pacing by limiting stops or adding buffers.	L	Optional	S2, S3

6. Functional Requirements

The 17 existing functional IDs, priorities and source assignments are summarized in Table 5-1. Each requirement below retains the nine detail fields required by the official template. Core and Optional status follows Chapter 5. Optional requirements retain their detailed definitions; their implementation and acceptance checks are not mandatory for the six-week MVP.

6.1. Functional Requirement REQ-1.1 — POI Search and Categories

Priority: H | Sources: S1,S2,S3,P | MVP Status: Core

6.1.1. Description

The system shall find Macau POIs by keyword and category.

6.1.2. System Input

Keyword, category, optional map area

6.1.3. Display

List/cards grouped by four categories, with map entry

6.1.4. System Processing

Filter curated records; allow clear/reset

6.1.5. System Output

Matching POIs and count

6.1.6. Other

Search works without a generated trip

Verification: A name search and a category filter return the expected curated POIs, and clearing filters restores the complete list.

6.1.7. Constraints

Approximately 30 curated POIs

6.1.8. Data Handling

Stable POI IDs and UTF-8 names

6.1.9. Error Handling

No matches: show empty state

6.2. Functional Requirement REQ-1.2 — POI Details and Opening Information

Priority: H | Sources: S1,S2,S3 | MVP Status: Core

6.2.1. Description

The system shall show practical visit information.

6.2.2. System Input

Selected POI ID

6.2.3. Display

Address/map, opening days/hours, last entry, suggested duration, known cost/booking note

6.2.4. System Processing

Read verified curated record; distinguish known and missing fields

6.2.5. System Output

POI detail page

6.2.6. Other

Opening information supports planning checks

Verification: A POI with a closed day, last-entry limit and missing price displays each condition correctly without inventing missing data.

6.2.7. Constraints

Not guaranteed real-time

6.2.8. Data Handling

Store coordinates, source/update notes when available

6.2.9. Error Handling

Missing data marked Unknown; no fabricated hours

6.3. Functional Requirement REQ-1.3 — Preference Match and Explanation

Priority: M | Sources: S3,P | MVP Status: Core

6.3.1. Description

The system shall show a Match Percentage and a short reason indicating preference alignment between the user's selected interests and a POI's categories or tags.

6.3.2. System Input

Selected interests and POI category/tags

6.3.3. Display

Match % and explanation

6.3.4. System Processing

Apply the existing simple preference-weighting rules and show a short explanation of matching categories or tags. Rank POIs within each category in descending Match order without excluding a selected POI solely because of a low score. The raw scoring formula need not be displayed.

6.3.5. System Output

Ranked POI suggestions

6.3.6. Other

Match is not itinerary feasibility or a measure of prediction accuracy for user satisfaction. It does not guarantee that a POI can be visited on a specified date or at a specified time.

Verification: Change interests and inspect the ranking and category/tag explanation. The interface shall not describe Match as itinerary feasibility, satisfaction-prediction accuracy or guaranteed access on a specified date/time. No complex machine-learning model or new scoring formula is required.

6.3.7. Constraints

No complex ML model required

6.3.8. Data Handling

Clamp percent to 0–100; persist tags

6.3.9. Error Handling

No preferences: show neutral ranking

6.4. Functional Requirement REQ-1.4 — Favorites and Add to Trip

Priority: H | Sources: S1,S2,S3,P | MVP Status: Core

6.4.1. Description

The system shall separate saving a POI from selecting it for a trip.

6.4.2. System Input

Favorite toggle; Add action; trip ID

6.4.3. Display

Distinct heart and add controls; selected status

6.4.4. System Processing

Update respective collections; prevent duplicate trip selection

6.4.5. System Output

Updated favorites/selected targets

6.4.6. Other

Favorites contain POIs, not whole shared trips

Verification: Favoriting a POI leaves trip targets unchanged; removing a trip target leaves its favorite unchanged.

6.4.7. Constraints

Adding a POI requires an active trip context. Removing it from a trip or deleting a trip shall not remove the independently saved favorite.

6.4.8. Data Handling

Use POI ID uniqueness per trip

6.4.9. Error Handling

Duplicate add prompts instead of creating second record

6.5. Functional Requirement REQ-1.5 — Trip Information Setup

Priority: H | Sources: S1,S3,P | MVP Status: Core

6.5.1. Description

The system shall capture trip constraints before planning.

6.5.2. System Input

Macau start, dates, per-day time windows, budget/currency, interests, selected POIs, named companions

6.5.3. Display

Editable per-day form and companion list

6.5.4. System Processing

Validate required values and date or time ranges. Reject an end time that is not later than its start time. Companion names populate bill payer and participant choices and shall not affect route generation.

6.5.5. System Output

Valid trip configuration

6.5.6. Other

Manual start point and companion names supported

Verification: Invalid dates or time ranges are rejected; changing only companion names leaves the same routing request unchanged.

6.5.7. Constraints

Companions do not influence itinerary routing

6.5.8. Data Handling

Store dates ISO format and times HH:mm

6.5.9. Error Handling

Missing/invalid inputs shown near fields

6.6. Functional Requirement REQ-1.6 — Automatic Feasible Itinerary

Priority: H | Sources: S1,S2,S3,P | MVP Status: Core

6.6.1. Description

The system shall generate ordered daily POI schedule minimizing unnecessary backtracking.

6.6.2. System Input

Trip settings, POIs, opening/visit/travel estimates

6.6.3. Display

Day tabs, stop order, start/end times and transfers

6.6.4. System Processing

Group selected POIs geographically and order them using the existing heuristic. Check the full visit against known opening windows, last entry, travel time and the daily end time. Compare estimated admission and transport costs with the budget and flag any excess. A must-visit preference may guide selection, but shall not override a known feasibility constraint.

6.6.5. System Output

Generated schedule and feasibility status

6.6.6. Other

Use heuristic rather than global optimality

Verification: A controlled set of POIs either fits known visit and travel windows or produces explicit unresolved constraints.

6.6.7. Constraints

Do not schedule closed or impossible visits as valid

6.6.8. Data Handling

Track durations and calculated arrival/departure times

6.6.9. Error Handling

No feasible plan: explain conflicts and preserve choices

6.7. Functional Requirement REQ-1.7 — Unscheduled POIs and Conflict Notices

Priority: H | Sources: S1,S3 | MVP Status: Core

6.7.1. Description

The system shall never silently discard selected POIs.

6.7.2. System Input

Proposed/edited schedule and constraints

6.7.3. Display

Affected POI, reason, unresolved list

6.7.4. System Processing

Detect closure, last-entry violation, day-end overflow and overlap

6.7.5. System Output

Warning and unscheduled list

6.7.6. Other

User decides whether to remove or reassign

Verification: An infeasible selected POI appears in the unscheduled list with a reason and is not silently removed.

6.7.7. Constraints

No automatic optimal substitute search

6.7.8. Data Handling

Store conflict code and POI reference

6.7.9. Error Handling

Unknown hours shown as uncertain, not confirmed closed

6.8. Functional Requirement REQ-1.8 — Itinerary Editing and Downstream Recalculation

Priority: H | Sources: S1,S2,S3,P | MVP Status: Core

6.8.1. Description

The system shall edit visit times and transport while rechecking later stops.

6.8.2. System Input

POI start and end times in 15-minute increments; transport mode selected manually by the user

6.8.3. Display

Updated downstream schedule, affected segments and alerts

6.8.4. System Processing

After a user changes visit times or manually selects a transport mode, recompute affected downstream POI arrival/departure times and travel segments. Recheck opening hours, last entry, time overlaps and the daily start/end time window. Update the map to match the revised order and display explicit conflict warnings. Never silently delete selected POIs. Timing recalculation may be automatic; deletion, replacement and overwriting still require confirmation under REQ-1.9.

6.8.5. System Output

Revised itinerary

6.8.6. Other

User retains control over major changes

Verification: Adjust POI start/end times in 15-minute increments and manually change a transport mode. Compare downstream times with the controlled fixture. Introduce each opening-hour, last-entry, overlap and daily-window conflict; verify explicit warnings and retention of every selected POI. Cancelling deletion, replacement or overwrite shall preserve the original data.

6.8.7. Constraints

Transport time estimates, not live timetables

6.8.8. Data Handling

Store change and updated timestamps

6.8.9. Error Handling

Conflict shows warning; do not silently delete stops

6.9. Functional Requirement REQ-1.9 — Deletion and Overwrite Confirmation

Priority: H | Sources: S3,P | MVP Status: Core

6.9.1. Description

The system shall confirm destructive itinerary actions.

6.9.2. System Input

Delete POI or replace saved plan request

6.9.3. Display

Confirmation dialog with affected trip name/stop

6.9.4. System Processing

Proceed only after explicit confirmation

6.9.5. System Output

Updated or unchanged plan

6.9.6. Other

Cancel must be reversible without mutation

Verification: Canceling a deletion or overwrite leaves the saved plan unchanged; confirming applies the intended change.

6.9.7. Constraints

Do not automatically replace selected POIs

6.9.8. Data Handling

Persist after confirmed action only

6.9.9. Error Handling

Cancel or storage failure preserves original

6.10. Functional Requirement REQ-1.10 — Map and External Navigation

Priority: H | Sources: S1,S2,S3,P | MVP Status: Core

6.10.1. Description

The system shall visualize itinerary route and allow leg-level navigation.

6.10.2. System Input

Trip/day or selected POI/leg

6.10.3. Display

Macau map, numbered markers, sequence and estimated modes/times

6.10.4. System Processing

Draw selected POIs; open navigation provider via explicit action

6.10.5. System Output

Map and navigation handoff

6.10.6. Other

Map is geographic, not a schematic diagram

Verification: Map marker order matches the itinerary; a failed map provider leaves the ordered text itinerary available.

6.10.7. Constraints

No live bus arrival or precise stop-side routing

6.10.8. Data Handling

Store coordinates and segment estimates

6.10.9. Error Handling

Provider unavailable: display text stop order

6.11. Functional Requirement REQ-1.11 — Macau Start Points and Landmarks

Priority: H | Sources: S1,S3,P | MVP Status: Core

6.11.1. Description

The system shall offer common Macau entry points and a manually specified Macau origin, and clearly identify major landmarks and entry or exit points on the map.

6.11.2. System Input

Chosen border crossing, airport or manual point

6.11.3. Display

Show an entry-point selector and map markers. Supported curated origins include Border Gate, Hengqin Port, the Hong Kong-Zhuhai-Macao Bridge Port, Macau International Airport, Taipa Passenger Terminal and Outer Harbour Passenger Terminal. Prominent landmark markers include the Ruins of St. Paul's, Grand Lisboa and Wynn Palace.

6.11.4. System Processing

Use selected point as first leg origin

6.11.5. System Output

Origin saved with trip

6.11.6. Other

Support first-time visitor orientation

Verification: A selected or valid manual Macau origin is saved and used for the first leg; entry points and landmark markers are identifiable.

6.11.7. Constraints

Only curated valid geographic points are geocoded

6.11.8. Data Handling

Store origin type and coordinate

6.11.9. Error Handling

Unrecognized manual point: request valid map location

6.12. Functional Requirement REQ-1.12 — Budget and Currency Reference

Priority: M | Sources: S1,S2,S3,P | MVP Status: Core

6.12.1. Description

The system shall show budget and MOP/HKD/CNY reference conversions.

6.12.2. System Input

Amount, selected currency, budget

6.12.3. Display

Show the converted reference amount, the applied rate and whether the rate is static or dated. Show its source and update time when available, together with the estimated budget comparison. A reference amount is not a guaranteed transaction rate.

6.12.4. System Processing

Apply available reference rate and show estimated budget comparison

6.12.5. System Output

Reference amount and comparison

6.12.6. Other

No actual payment service

Verification: A known amount converts using the displayed reference rate; a missing rate produces an unavailable or explicitly dated fallback state.

6.12.7. Constraints

FX may be delayed or static; label clearly

6.12.8. Data Handling

Use decimal currency values and round display

6.12.9. Error Handling

Missing rate: show unavailable or clearly dated fallback

6.13. Functional Requirement REQ-1.13 — Travel Bill and Equal Expense Splitting

Priority: M | Sources: S3,P; S1 non-essential; S2 low ranking | MVP Status: Optional

6.13.1. Description

If this Optional capability is implemented, the system shall record trip expenses and compute equal shares.

6.13.2. System Input

Place, payer, amount, MOP/HKD/CNY, participant subset

6.13.3. Display

Bill overview; Add Bill; separate Settlement view

6.13.4. System Processing

Convert each recorded expense to CNY using its retained reference rate and divide it equally among its selected participants. Compute each person's paid amount, required share and net transfer balance. Reconcile rounding so the displayed shares add to the recorded expense; non-participants shall bear no share.

6.13.5. System Output

Expense records and CNY settlement

6.13.6. Other

Companions manually named; independent of planner

Verification: Selected participants alone share each bill; CNY paid amounts, shares and net transfers reconcile after rounding.

6.13.7. Constraints

No complex weighted split or real payments

6.13.8. Data Handling

Validate positive amounts and nonempty participant set

6.13.9. Error Handling

Invalid payer/participants/currency: block saving

6.14. Functional Requirement REQ-1.14 — Save and Reopen Trips

Priority: H | Sources: S3,P | MVP Status: Core

6.14.1. Description

The system shall allow retrieval of previously generated and edited trips.

6.14.2. System Input

Save/open trip action and trip ID

6.14.3. Display

My Trips list and saved itinerary details

6.14.4. System Processing

Persist trip configuration, POIs, schedule and edits

6.14.5. System Output

Restored itinerary

6.14.6. Other

Mine is one of the four persistent bottom tabs; My Macau Trips is a functional page accessible from Home or Mine.

Verification: Reopening a saved trip restores its inputs, stop IDs, order and accepted time edits; corrupt data produces a recoverable message.

6.14.7. Constraints

Avoid unintentional overwrite

6.14.8. Data Handling

Use stable trip IDs and date serialization

6.14.9. Error Handling

Missing/corrupt trip: show recoverable message

6.15. Functional Requirement REQ-1.15 — Itinerary Plaza and Sharing

Priority: M | Sources: S1,S3,P | MVP Status: Core

6.15.1. Description

The system shall browse and voluntarily publish Macau itineraries.

6.15.2. System Input

Publish request, shared post, favorite POI

6.15.3. Display

Discover/Plaza feed and post itinerary detail

6.15.4. System Processing

Create public post from selected trip; allow individual POI favorites

6.15.5. System Output

Shared post and POI favorites

6.15.6. Other

No copy/import of entire other users trips

Verification: The publish preview excludes private companions, bills and budgets; a shared post permits saving one POI without whole-trip import.

6.15.7. Constraints

Sharing must be opt-in; no real-time collaboration

6.15.8. Data Handling

Exclude companion names, bill details, personal budgets and private notes from public posts. Display the proposed public content for explicit confirmation before publishing.

6.15.9. Error Handling

Missing post or publish failure: show clear error

6.16. Functional Requirement REQ-1.16 — Nearby Food Suggestions

Priority: M | Sources: S2,S3; team scope decision | MVP Status: Optional

6.16.1. Description

If this Optional capability is implemented, the system shall suggest curated food POIs near selected stops.

6.16.2. System Input

Selected stop or itinerary region

6.16.3. Display

Nearby food cards with distance estimate

6.16.4. System Processing

Filter food category in existing POI dataset and sort by proximity

6.16.5. System Output

Optional food suggestions; add only by user action

6.16.6. Other

No automatic restaurant booking or live review scraping

Verification: Suggestions come only from nearby curated food POIs, and none is added to a trip without a user action.

6.16.7. Constraints

Limited to curated POIs

6.16.8. Data Handling

Reuse POI coordinates and category

6.16.9. Error Handling

No suitable food POIs: show no recommendations

6.17. Functional Requirement REQ-1.17 — Relaxed Pace Option

Priority: L | Sources: S2,S3; optional enhancement | MVP Status: Optional

6.17.1. Description

If this Optional capability is implemented, the system shall offer simpler pacing by limiting stops or adding buffers.

6.17.2. System Input

Standard/relaxed choice or daily stop cap

6.17.3. Display

Pace option and resulting day schedule

6.17.4. System Processing

Adjust stop cap/buffer before heuristic scheduling

6.17.5. System Output

Less dense itinerary with unscheduled list

6.17.6. Other

Enhancement after core planner stability

Verification: Relaxed mode applies its configured stop cap or buffer and preserves every unallocated selection in the unscheduled list.

6.17.7. Constraints

No exact walking-distance optimization

6.17.8. Data Handling

Store pace mode and cap

6.17.9. Error Handling

Too many selected stops: preserve unscheduled list

7. External Interface Requirement

7.1. Data Interfaces

POI records use stable IDs and include category, Macau coordinates, opening days or hours, applicable last-entry time, suggested visit duration and known price or reservation notes. Unknown values shall remain distinguishable from zero or free admission. Store the available information source and update note. Trip, favorite, post and bill records use stable references, UTF-8 text, structured dates and currency codes. Companion data supplies bill choices but is excluded from planner inputs.

7.2. User Interfaces

Responsive four-tab navigation; four home shortcuts; POI cards with separate Favorite and Add; day-based itinerary with per-stop actions (Edit Time, Add Bill, Delete); bottom itinerary actions Share, Replan, Map.

7.3. Other Interfaces

Leaflet and OpenStreetMap provide map presentation under the working technical plan. A user action may open Google Maps or an equivalent external navigation provider. Exchange-rate and weather feeds are optional reference interfaces and shall expose unavailable or dated data clearly. The MVP has no live public-transport, booking, payment or social-scraping integration.

8. Non-Functional Requirements

8.1. System Performance

Table 8-1 System performance requirements

Req. ID	Required behavior	Verification and target status
PERF-1.1	Navigation and controls remain usable and give visible feedback while work is in progress.	Measure from a user action to visible feedback with browser timestamps or a time-stamped recording. Exercise navigation and busy states; every trial must retain usable controls and show the expected feedback. A latency limit is pending approval.
PERF-1.2	Page and POI-list loading show progress when delayed and end in a result, empty state or actionable error.	Use the 30-POI fixture for populated, empty-result and failed-load scenarios. Measure action-to-usable-list time; record cold and warm cache trials separately. Pass the expected result/empty/error checks; the loading-time limit is pending approval.
PERF-1.3	Itinerary generation completes with a schedule or an actionable error without losing selected POIs.	Use 5- and 10-selected-POI cases from the 30-POI fixture, including infeasible opening/last-entry and daily-window cases. Measure Generate-to-rendered-result/error time. Every selected ID must remain scheduled or explicitly unscheduled. The generation-time limit is pending approval.

Req. ID	Required behavior	Verification and target status
PERF-1.4	A delayed optional external provider does not freeze the whole application.	Stub an unavailable or delayed external provider. Measure request-to-timeout/error feedback; verify responsive controls and access to the core trip. A missing warning, frozen workflow or lost selections fails. The timeout duration is pending approval.

The acceptance record shall identify the device model, CPU/RAM, OS and browser versions, application build, viewport, cache state, network profile and provider/stub state. Use a versioned fixture of 30 Macau POIs with named IDs, verified opening/last-entry data and recorded travel-time assumptions. Record the selected IDs and expected schedule, warnings or error state for each scenario.

Run 10 recorded trials per agreed scenario and cache condition; report individual timings, median, 95th percentile and maximum, with the calculation method. A functional check passes only when all trials produce the expected fixture outcome without lost selections or cross-trip corruption. Numerical timing/load acceptance remains Not Evaluated until limits, devices and conditions are approved; after approval, an exceeded limit fails. Proposed Targets in the revision report are unmeasured discussion values, not system commitments.

8.2. Information Security

Table 8-2 Information security requirements

Req. ID	Requirement
SEC-1.1	Private trips, companion names and bills shall not be public by default.
SEC-1.2	Public Plaza posts shall exclude companion names, bill details, personal budgets and private notes.
SEC-1.3	Sharing or publishing shall require an explicit user action and confirmation of the proposed public content.
SEC-1.4	Inputs shall be validated, and private data shall not be exposed through public views or exports.

8.3. Availability

Table 8-3 Availability requirements

Req. ID	Description
AVA-1.1	Core trip viewing should remain usable when optional FX/weather feed is unavailable
AVA-1.2	Map failure provides POI text list and route order
AVA-1.3	No guarantee of offline maps or continuous third-party uptime

8.4. Capacity

Table 8-4 Capacity requirements

Req. ID	Description
CAP-1.1	Use the versioned 30-POI acceptance fixture. Record dataset version/count and verify search, details, planning and recalculation against expected fixture outputs. A wrong or missing result fails; testing this fixture is not a claim of maximum dataset capacity.

Req. ID	Description
CAP-1.2	Create, save and reopen multiple distinct trips in one test profile. Record trip count, storage mechanism and browser quota; compare IDs, POI order and edits before/after reload. Pass only with correct separation and no overwrite or data loss. Agree the test trip count before acceptance.
CAP-2.1	Concurrent load is not yet confirmed. Record deployment mode, device/server resources, browser sessions, simultaneous operations, duration, errors and timings. Approve load levels and limits before a capacity pass/fail claim; exploratory results alone do not establish supported concurrency.

9. Domain Requirements

Table 9-1 Macau domain requirements

ID	Requirement	Source
DR-01	Macau border crossings/airport as origins	S1,S3,P
DR-02	Opening days/hours, last entry and visit duration	S1,S2,S3
DR-03	Walking/bus and locally applicable estimated travel	S1,S2,S3
DR-04	MOP/HKD/CNY reference conversions	S1,S2,S3
DR-05	Actual Macau coordinates and landmarks	S1,S3
DR-06	History/culture, food, shopping, nature categories	S1,P

Appendix A Requirements Elicitation and Consolidation

S1 is the assigned-customer interview with D2-Team1 (ChabataMUST). S2 contains the potential-user questionnaire and its reported 94 responses. S3 is the consolidated summary attributed to four development-team members as users. These sources support different parts of the requirements; a source citation does not imply that the assigned customer approved every S2 or S3 addition. P is the earlier project baseline, not a fourth elicitation population.

The consolidated source mapping appears in Table A-5. The matrix distinguishes supporting evidence from final scope approval.

A.1. Source Identification and Collection

Table A-1 Requirements sources and evidence limits

Source	Method and evidence	Limits
S1 Assigned customer	ChabataMUST interview, 10 October 2026. [S1], original and expanded answers Q1-Q10.	Expanded wording is reported as confirmed by the development team. The customer group is D2-Team1 (ChabataMUST); a separate confirmation record is not independently supplied.
S2 Potential users	Questionnaire, six-page platform summary and Q14-Q16 open-answer exports. 94 reported responses, WeChat, 13-15 October 2026.	Counts describe the available respondent sample. Multi-select shares are not additive. Some respondents lacked Macau travel experience.
S3 Team members	Consolidated summary version 2 dated 13 October 2026, covering 11 questions and attributions to Members 1-4.	Four individual completed questionnaires are not supplied; counts are reported from the summary and are not generalizable to all tourists.
P Project baseline	Final Task 1 Proposal, D2-Team2, 8 October 2026.	Prior scope and design decisions; not an independent elicitation population or proof of customer approval.

A.2. Source 1 — Confirmed Customer Needs

Table A-2 Assigned customer needs and requirement mapping

Customer topic	Requirement reflected in this URS	Source and related IDs
POI access	Search by name or category, map browsing and saving individual POIs. Separation of Favorite and Add is additionally supported by S2/S3/P.	S1 Q3, Q10; REQ-1.1, 1.2, 1.4
Planning	Reduce unnecessary distance and travel time, consider budget and opening hours, and explain unusable visits.	S1 Q2, Q4; REQ-1.5-1.7
Adjustments	Change visit time or transport mode and recalculate later travel or arrival times with conflict warnings.	S1 Q5; REQ-1.8

Customer topic	Requirement reflected in this URS	Source and related IDs
Macau orientation	Map routes, applicable transport options, entry and exit points and prominent landmarks.	S1 Q1, Q5, Q9, Q10; REQ-1.10-1.11
Currencies	Reference conversion among MOP, HKD and CNY.	S1 Q6, Q9; REQ-1.12
Plaza	Browse and publish itineraries and save individual POIs from shared content.	S1 Q7; REQ-1.15
Bill boundary	The assigned customer considers full bill splitting non-essential. Optional manual equal splitting is retained from P and S3.	S1 Q6; REQ-1.13 (M, optional)

A.3. Source 2 — Survey Findings

Counts and percentages below are checked against the platform summary [S2-R]. The denominator is 94 for each displayed item. Percentages are recomputed from counts and rounded to two decimal places; multi-select selections do not total 100%. The three open-answer exports contain matching sets of 94 sequence IDs and submission timestamps. This confirms record alignment, not that every open answer is substantive or that the sample represents all Macau travelers.

Table A-3 Checked potential user survey findings

Finding	Checked counts and percentages	Implication and evidence
Travel style	76/94 (80.85%) independent travel	Self-service trip workflow. S2-R p.1 Q2.
Pain points	54/94 (57.45%) backtracking; 52/94 (55.32%) insufficient time	Often or sometimes in the full sample. REQ-1.6-1.7. S2-R p.2 Q7.
Planning goals	73/94 (77.66%) less backtracking; 61/94 (64.89%) relaxed travel	Core routing and lower-priority pace enhancement. REQ-1.6, 1.17. S2-R p.3 Q8.
Change handling	52/94 (55.32%) warnings with user choice; 29/94 (30.85%) automatic shifting and warnings	Recalculate time but retain user control over destructive changes. REQ-1.8-1.9. S2-R p.2 Q5.
Saving POIs	49 favoriting + 29 both actions = 78/94 (82.98%) need favorites	Favorite and Add remain separate. REQ-1.4. S2-R p.2 Q6.
Trip display	63/94 (67.02%) transport; 55/94 (58.51%) map; 49/94 (52.13%) cost details	Transport segments, maps and cost reference. REQ-1.10, 1.12. S2-R p.4 Q12.
Additional features	66/94 (70.21%) food; 38/94 (40.43%) FX; 36/94 (38.30%) favorites; 14/94 (14.89%) group bills	Retain food, currency and favorites at their stated priority; optional simple bills. S2-R p.5 Q13.
Pace and sharing	31/94 (32.98%) relaxed mode; 16/94 (17.02%) sharing	Optional or phased enhancements. REQ-1.17, 1.15. S2-R p.5 Q13.
Quality	71/94 (75.53%) ease; 60/94 (63.83%) accuracy; 42/94 (44.68%) clear maps	Usability and data checks; these counts imply no 2-second or 95% guarantee. S2-R p.3 Q10.

The platform summary records 25 of 94 respondents (26.60%) as having visited Macau, and 42 of 94 (44.68%) as having a confirmed trip or an intention to visit. Some respondents lacked Macau travel experience. The experience-question proportions therefore describe this full respondent sample and shall not be interpreted as representative incidence rates among experienced visitors.

A.4. Source 3 — Team-Member Findings

Table A-4 Reported team member findings

Topic	Reported member feedback	Mapped requirement
Core priorities	Planning 4/4; map/navigation 3/4; editing/recalculation 3/4 in top-three choices. Search and setup each receive one top-three vote.	S3 Q10; REQ-1.6, 1.10, 1.8, 1.1, 1.5
Feasibility	4/4 support an unscheduled list with reasons and no silent loss of selected POIs.	S3 Q03; REQ-1.7
User control	4/4 support confirmation before deleting, replacing or overwriting a plan.	S3 Q04; REQ-1.9
POI actions	4/4 distinguish Favorite from Add and ask for a Match explanation rather than a feasibility guarantee.	S3 Q06-Q07; REQ-1.3-1.4
Expenses and privacy	The summary reports support for simple manual three-currency bills and private companion/bill information. Its detailed privacy attribution specifically names Members 1, 2 and 4.	S3 Q08-Q09 and consolidated list; REQ-1.13, SEC-1.1-1.3
Enhancements	Members 3 and 4 support lighter pacing. Member 3 prefers preserving the entered order; the consolidated summary also recommends a keep-order option.	S3 Q03 and conclusions; REQ-1.17 (L). Manual-order lock remains deferred in the existing URS.

A.5. Consolidated Traceability Matrix

Table A-5 Requirements traceability matrix

Req. ID	Requirement	S1 interview	S2 survey	S3 summary	Decision
REQ-1.1	POI Search and Categories	Q3, Q10: search and categories	Q11: practical POI information	Q01-Q02: integrated discovery	H; retain.
REQ-1.2	POI Details and Opening Information	Q4, Q8-Q9: reliable hours	Q11: hours, price, last entry	Q02: last entry and duration	H; retain.
REQ-1.3	Preference Match and Explanation	-	No explicit Match requirement	Q07: explanation and limits	M; S3 and P baseline.
REQ-1.4	Favorites and Add to Trip	Q3, Q7, Q10: save a POI	Q6: separate save/add choices	Q06: independent actions	H; retain.
REQ-1.5	Trip Information Setup	Q2: budget, duration, origin	Q5, Q8: time/planning context	Q03 and Q08: constraints and companions	H; S1, S3 and P baseline.
REQ-1.6	Automatic Feasible Itinerary	Q2, Q4: distance/time/hours	Q8: 73/94 less backtracking	Q03, Q10: 4/4 planning	H; core; P baseline.

Req. ID	Requirement	S1 interview	S2 survey	S3 summary	Decision
REQ-1.7	Unscheduled POIs and Conflict Notices	Q4: explain closed visits	Q7: time/opening pain points	Q03: 4/4 preserve choices	H; retain.
REQ-1.8	Itinerary Editing and Downstream Recalculation	Q5: edit and recalculate	Q5 and Q13: change handling	Q04, Q10: 3/4 top-three	H; core; P baseline.
REQ-1.9	Deletion and Overwrite Confirmation	-	Q5 supports user choice, not explicit deletion approval	Q04: explicit confirmation	H; S3 and P baseline.
REQ-1.10	Map and External Navigation	Q1, Q5, Q9-Q10: navigation	Q12: 63/94 transport, 55/94 map	Q05, Q10: map and navigation	H; core; P baseline.
REQ-1.11	Macau Start Points and Landmarks	Q9-Q10: entries and landmarks	No explicit border-marker requirement	Q05: origin-to-first-stop	H; S1, S3 and P baseline.
REQ-1.12	Budget and Currency Reference	Q6, Q9: MOP/HKD/CNY	Q13: 38/94; Q15.1 / export Q15	Q08: common settlement currency	M; retain; reference only.
REQ-1.13	Travel Bill and Equal Expense Splitting	Q6: full splitting non-essential	Q13: 14/94; Q15.2 / export Q16	Q08: simple bills, selected participants	M; Optional; Project Leader decision.
REQ-1.14	Save and Reopen Trips	-	-	Q04 and Q10: saved plans; may be phased	H; retain P baseline.
REQ-1.15	Itinerary Plaza and Sharing	Q7: browse and publish	Q13: 16/94 select sharing	Q09: privacy, individual POI saves	M; retain; limited Plaza in P.
REQ-1.16	Nearby Food Suggestions	-	Q13: 66/94 select food	Q02, Q09: food context	M; Optional; Project Leader decision.
REQ-1.17	Relaxed Pace Option	-	Q8: 61/94 relaxed goal; Q13: 31/94 mode	Members 3-4: lighter pace	L; Optional; Project Leader decision.

A dash means no explicit supporting evidence was identified for that source; it does not mean rejection. Source references identify supporting topics, not identical wording or unanimous approval. P is the project baseline. The original H/M/L priorities and optional-feature decisions are retained; final scope approval is still pending.

A.6. Conflicts and Resolution

Table A-6 Requirement conflicts and scope resolutions

Issue	Evidence	Retained decision and rationale
Core focus	S1 emphasizes search/navigation; S2 prioritizes less backtracking; S3 prioritizes planning.	Retain search, maps and planning as H. Integrate data and search before planner validation.

Issue	Evidence	Retained decision and rationale
Group bills	S1 Q6 considers full splitting non-essential; S2 Q13 selects it for 14/94; S3 and P support simple bills.	Project Leader decision, 18 October 2026: REQ-1.13 is M / Optional. Task 1 and the frontend planned Travel Bill, but S1 feedback and the low S2 ranking justify excluding it from mandatory six-week acceptance. Retain manual equal-share design; no payments or weighted splitting. This is not a claim of approval by all four members.
Ordering	S3 includes a preference to keep entered order; P and the existing URS use automatic ordering.	Retain automatic ordering, preserve selected POIs and request confirmation for major changes. A manual-order lock remains deferred and requires team scope agreement.
Food and pacing	S2 Q13 selects food for 66/94 and relaxed mode for 31/94; S3 adds meal/rest and pacing scenarios.	Project Leader decision, 18 October 2026: REQ-1.16 is M / Optional and REQ-1.17 is L / Optional. Retain curated-food and relaxed-pace definitions/designs; implement only if time permits, outside mandatory six-week MVP acceptance.
Numeric quality targets	S2-R asks for qualitative speed and accuracy. S2-S proposes 2-second, 95%, completeness and refresh figures.	Retain the qualitative quality requirements. Use the Chapter 8 test records and fixtures; numerical limits remain unapproved. Evaluate the revision report Proposed Targets by measurement before agreement; do not treat them as customer guarantees.
External data	S2-S proposes live FX/weather; P does not assume live FX and excludes weather-aware replanning and live transit.	Retain dated or static currency reference and optional informative weather. Live integration and weather-aware replanning are not mandatory MVP scope.

A.7. Evidence Retention and Pending Confirmation

Retain the supplied S1 interview PDF, the S2 questionnaire, platform report and open-answer exports, and the S3 consolidated summary as supporting project evidence. The individual S3 questionnaires, the two underlying S1 clarification DOCX files and a separate confirmation record are not included in the supplied evidence set. The member counts are therefore reported from S3 rather than independently checked against individual forms.

The assigned customer group is confirmed as D2-Team1 (ChabataMUST). On 18 October 2026, the Project Leader confirmed REQ-1.13 (M), REQ-1.16 (M) and REQ-1.17 (L) as Optional for the six-week MVP. Their definitions and design remain retained, but implementation is not mandatory for current MVP acceptance. These three scope decisions are no longer pending. Numerical performance/load limits and their test conditions still require measurement and agreement; this decision does not assert formal approval by all four team members.

Appendix B Scope Decisions

Retained and optional capabilities are summarized in Table B-1. Deferred capabilities follow in the next table. Core is mandatory for six-week MVP acceptance. Optional retains requirement definition and design but is implemented only if time permits and is not a mandatory acceptance function. Deferred is not implemented in this MVP. Priority expresses importance and does not determine MVP Status.

The Project Leader confirmed the three Optional scope decisions on 18 October 2026 based on requirements analysis and 's review comments. Travel Bill was planned in Task 1 and the frontend; making REQ-1.13 Optional is an explicit scope adjustment based on customer feedback and survey findings. This records the Project Leader's decision, not formal approval by all four members. Other Core classifications follow the retained Task 1 baseline and the existing scope definition, including lightweight Plaza and static/dated currency reference; live feeds and full social collaboration are not thereby made Core.

Table B-1 Retained and optional scope

Capability	MVP status and retained scope	Basis
Search, POI details, separate Favorite/Add, trip inputs, planning, warnings, edits, maps, origins and saved trips	Core. Required for current six-week acceptance; preserve existing H/M priorities. Currency reference may use static/dated rates; no live-rate service is promised.	REQ-1.1-1.12 and 1.14, with reference currency at M
Limited Plaza and opt-in sharing	Core, limited to the Task 1 lightweight Plaza: browse/publish, opt-in sharing and individual-POI saves. No whole-trip copying, full social network or real-time collaboration.	REQ-1.15; S1 Q7, S3 and P
Manual bills and equal CNY settlement	Optional / M. Project Leader confirmed; retain bill and equal-share design, implement only if time permits. Not mandatory for six-week MVP acceptance.	REQ-1.13; S1 Q6, S3 and P; narrowed Task 1 commitment
Nearby curated food suggestions	Optional / M. Project Leader confirmed; curated food suggestions remain defined/designed but are not mandatory for six-week MVP acceptance.	REQ-1.16; S2 and S3
Relaxed pace	Optional / L. Project Leader confirmed; retain lighter pacing design and implement only if time permits, after core stability.	REQ-1.17; S2 and Members 3-4
Weather and exchange-rate feeds	Optional external feeds. Core REQ-1.12 may use static/dated reference data; no mandatory live integration. Weather-aware automatic replanning remains Deferred.	Existing URS, P and source conflict resolution

Table B-2 Deferred capabilities and MVP alternatives

Deferred	Reason	MVP alternative
Live bus arrivals	Live API/data dependency	Estimated time + external navigation
Real-time collaboration	Sync and permissions	Share read-only itinerary

Deferred	Reason	MVP alternative
Exact walking optimization	Complex routing	Region grouping heuristic
Automatic expense import	Financial integration	Manual expense entry
Auto best substitute POIs	Complex replanning	Warnings and unscheduled list
Offline maps and social scraping	Data/cache and platform constraints	Online map and manual POI selection

Appendix C — Team Contributions

( ) — **Project Leader.** As the project leader, I coordinated team activities, monitored project progress, and facilitated communication among team members. My functional responsibilities covered Trip Information, User Preferences, Companion Data, and Travel Bill and Expense Settlement. For Task 2, I contributed to requirements analysis and consolidation, coordinated the preparation and integration of the User Requirements Specification (URS), and reviewed the document for consistency, completeness, formatting, and compliance with the assignment requirements.

( ) — **Requirements Elicitation and Initial URS Drafting.** was primarily responsible for customer requirements elicitation (Source 1) and contributed to the design and revision of questionnaires for Sources 2 and 3. helped collect and clarify customer requirements from D2-Team1, consolidate findings from different requirements sources, and prepare the initial URS draft.

( ) — **User Survey, Data Analysis, and Interface Requirements.** was responsible for designing and distributing the potential-user questionnaire (Source 2), collecting and cleaning 94 valid responses, and conducting quantitative analysis to identify key user needs and functional priorities. As the data and API service lead (Module C), also contributed to defining the POI data schema, C-to-D and C-to-B interface specifications, and data validation criteria using five representative test attractions to support subsequent system development and integration.

( ) — **Team Requirements Elicitation and Survey Recruitment.** contributed to collecting and clarifying requirements from development team members (Source 3). also assisted in recruiting potential users to participate in the Source 2 questionnaire, supporting the collection of user feedback for requirements analysis.